

COMP 4423: DEEP IMAGE RETRIEVAL

Easy Computer Vision

Xiaoyong Wei (魏驍勇)
x1wei@polyu.edu.hk

The Hong Kong Polytechnic University
香港理工大學
Opening Minds • Shaping the Future
啟迪思維 • 成就未來



Outline

- Deep image retrieval
- Feature aggregation/embedding/fusion
- Fine tuning (Siamese/Triplet networks)

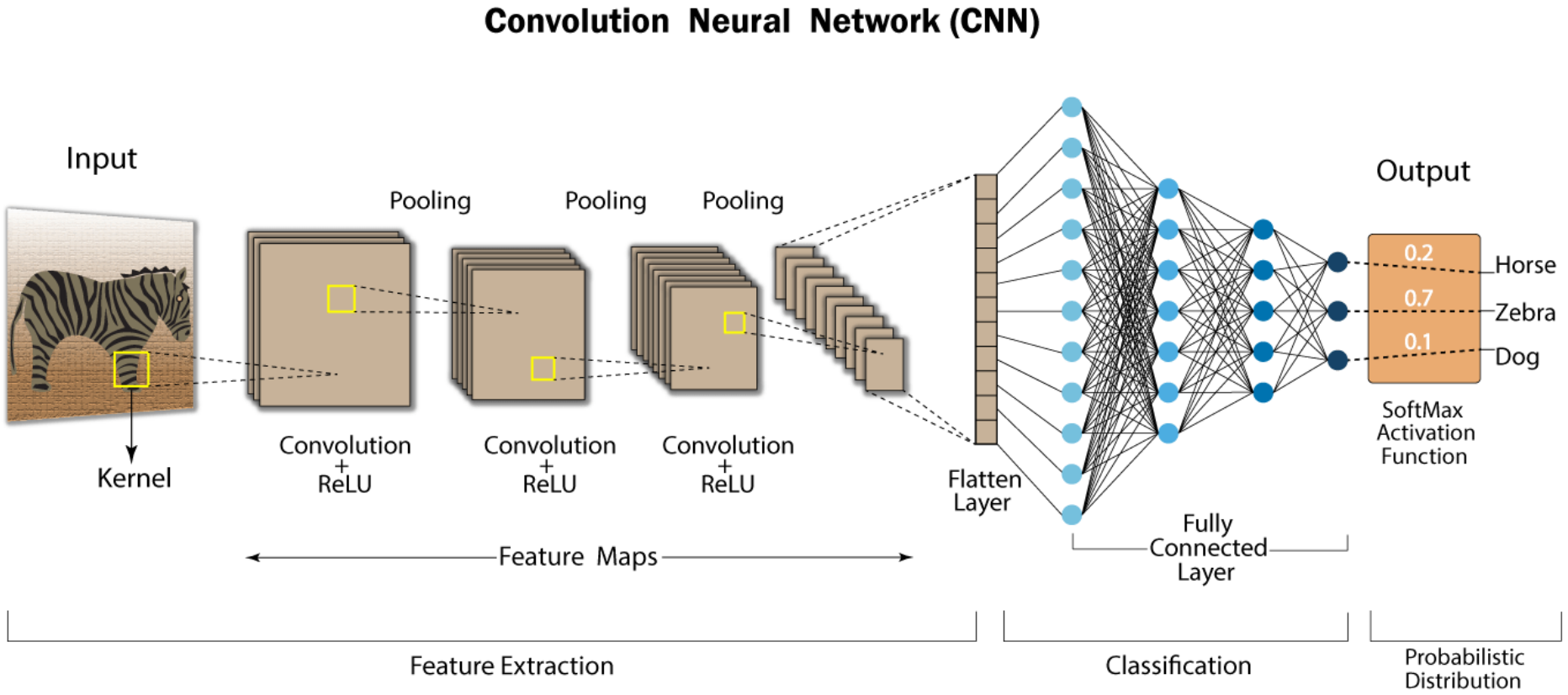


Deep Learning is cool. It’s in fact a game changer not only for classification, but also a wide range of Computer Vision tasks.

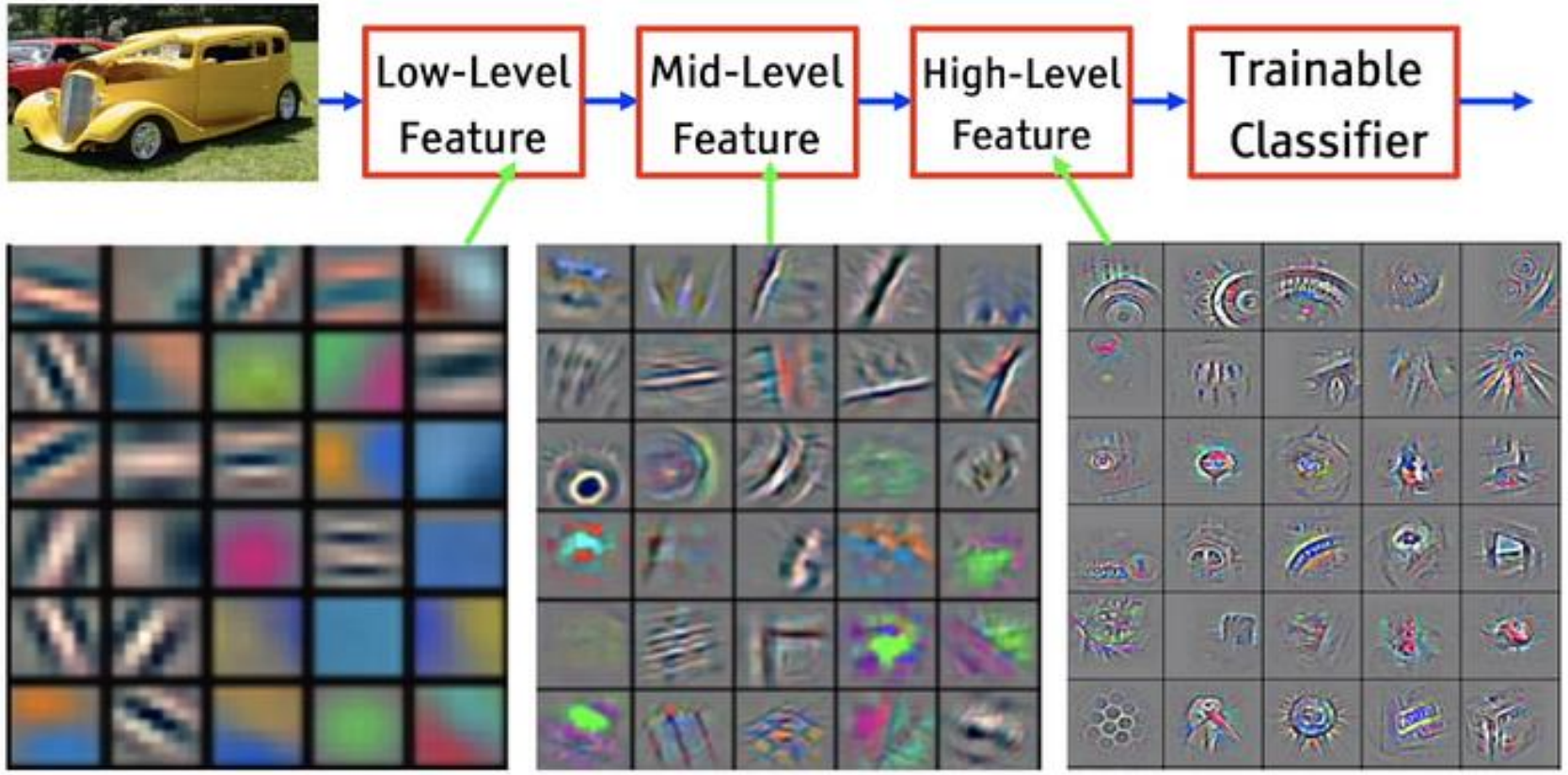


Let’s see how it helps the image retrieval

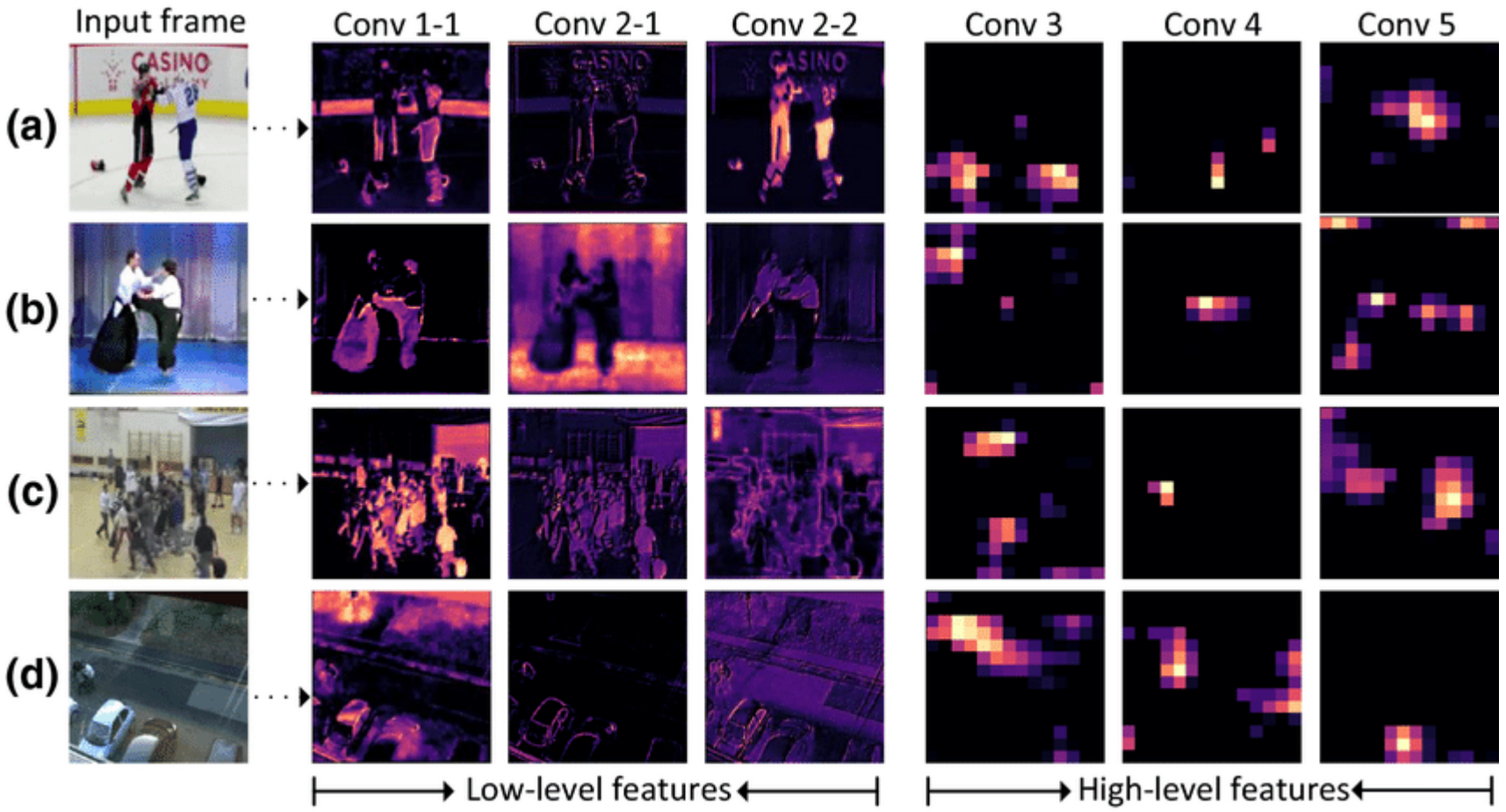
Convolutional Networks



https://discuss.boardinfinity.com/t/what-do-you-mean-by-convolutional-neural-network/8533

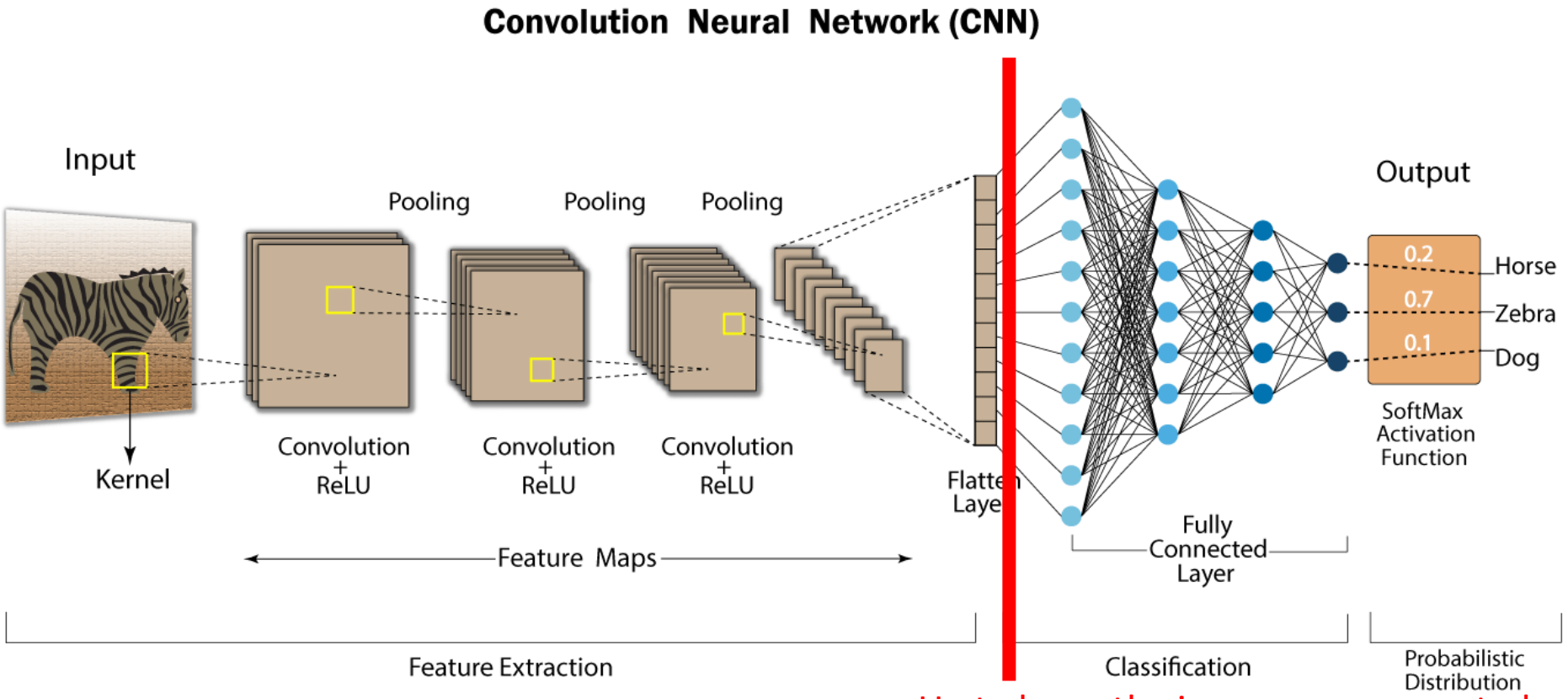


https://medium.com/analytics-vidhya/the-world-through-the-eyes-of-cnn-5a52c034dbeb



https://www.researchgate.net/publication/342428683_Multi-frame_feature-fusion-based_model_for_violence_detection?_tp=eyJjb250ZXh0Ijp7ImZpcnN0UGFnZSI6Il9kaXJlY3QiLCJwYWdlIjoiX2RpcmVjdCJ9fQ

Convolutional Networks



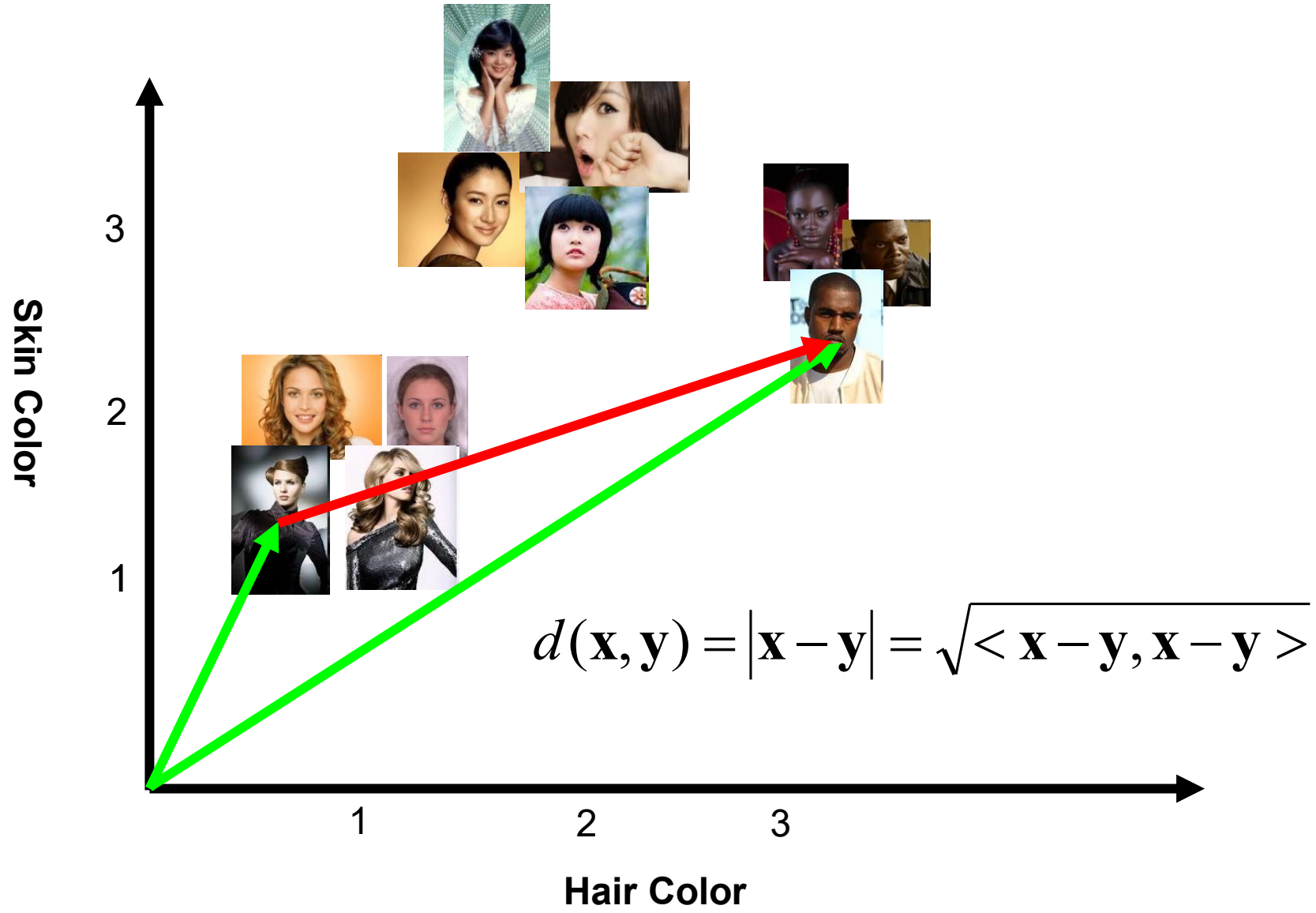
Up to here, the images are converted into feature vectors (represented in the feature space).

https://discuss.boardinfinity.com/t/what-do-you-mean-by-convolutional-neural-network/8533



As mentioned early, as long as the images are represented in the feature space, the search can be conducted by ranking images using similarity/distance metrics.

Metrics – Euclidian Distance

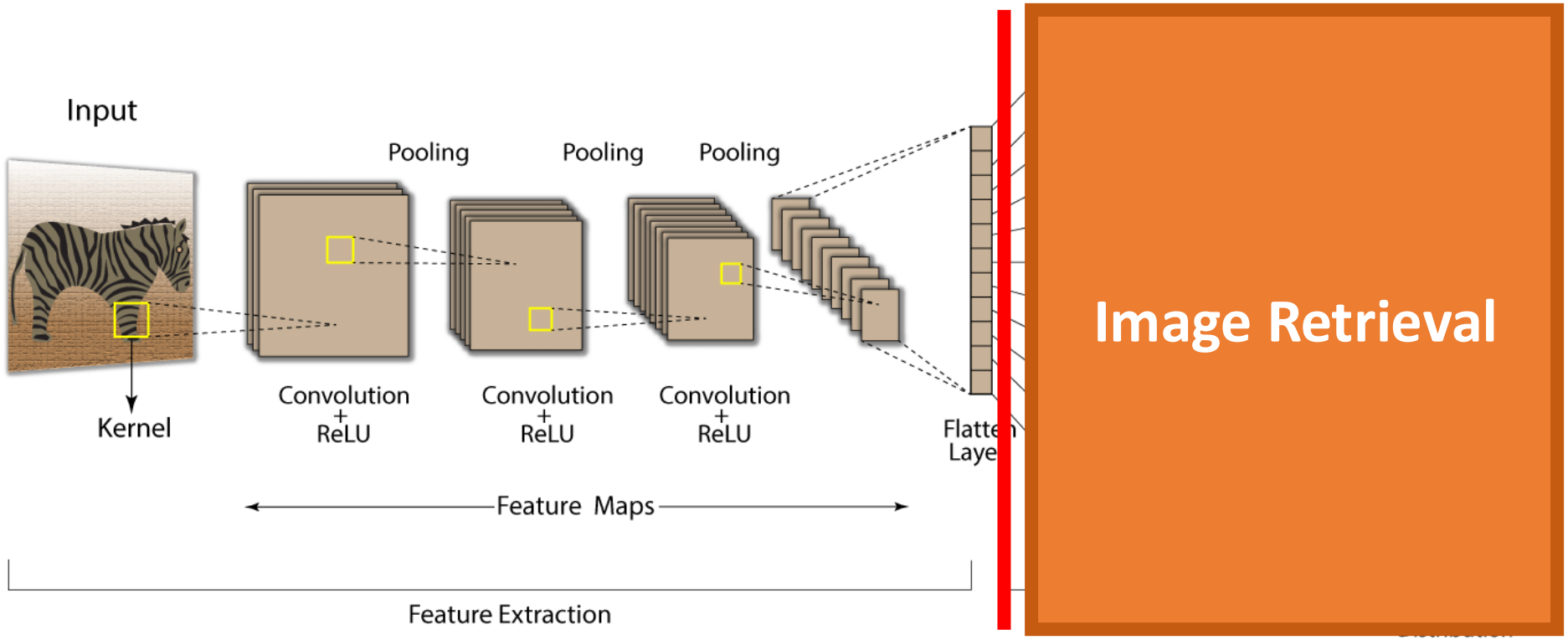


$$d(\mathbf{x},\mathbf{y}) = |\mathbf{x}-\mathbf{y}| = \sqrt{<\mathbf{x}-\mathbf{y},\mathbf{x}-\mathbf{y}>}$$



The only difference is that the feature space is now spanned by **deep** feature vectors.

Deep Image Retrieval



Up to here, the images are converted into feature vectors (represented in the feature space).

https://discuss.boardinfinity.com/t/what-do-you-mean-by-convolutional-neural-network/8533

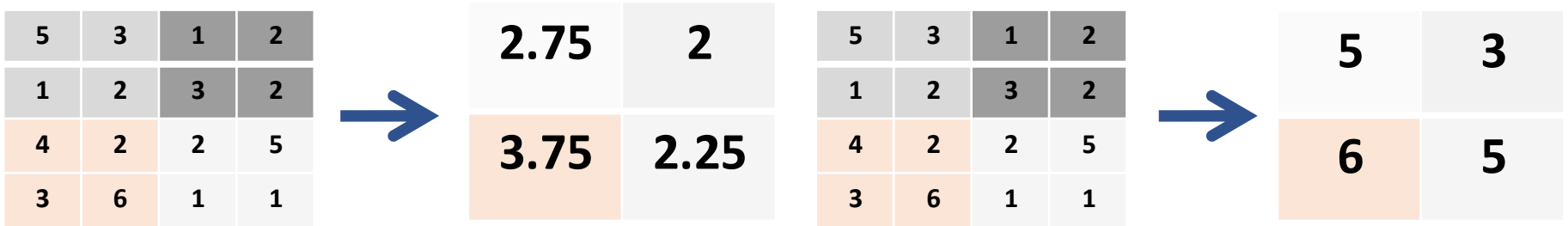


What’s the best way of using deep features?

Can we construct better features instead of using the raw feature maps?

Feature Aggregation

In feature maps the spatial dimensions of the original images are “preserved”. We can thus **summarize** the features over the spatial dimensions for better representations of regions. This can be done by using different types of pooling algorithms.

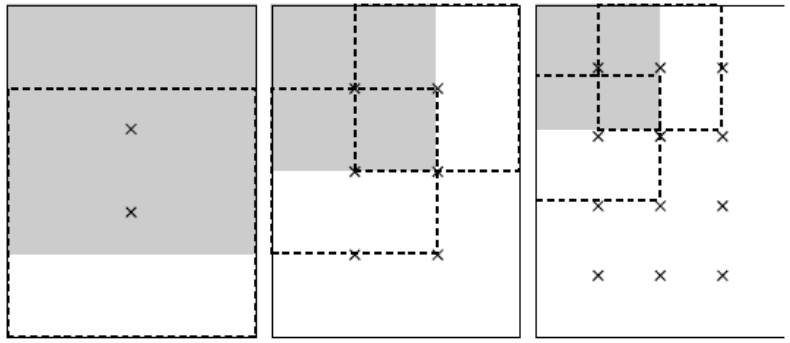


Sum/average Pooling **Max Pooling**

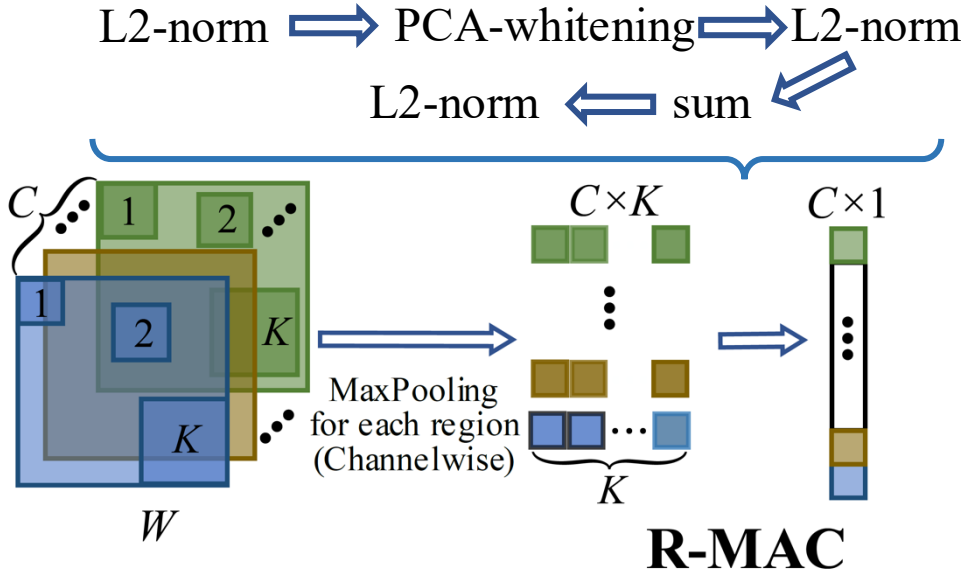
Chen W, Liu Y, Wang W, et al. Deep image retrieval: A survey[J]. arXiv preprint arXiv:2101.11282, 2021.

Single Feed - Forward Pass

R-MAC derives a compact image representation from the convolutional layers to encode multiple image regions

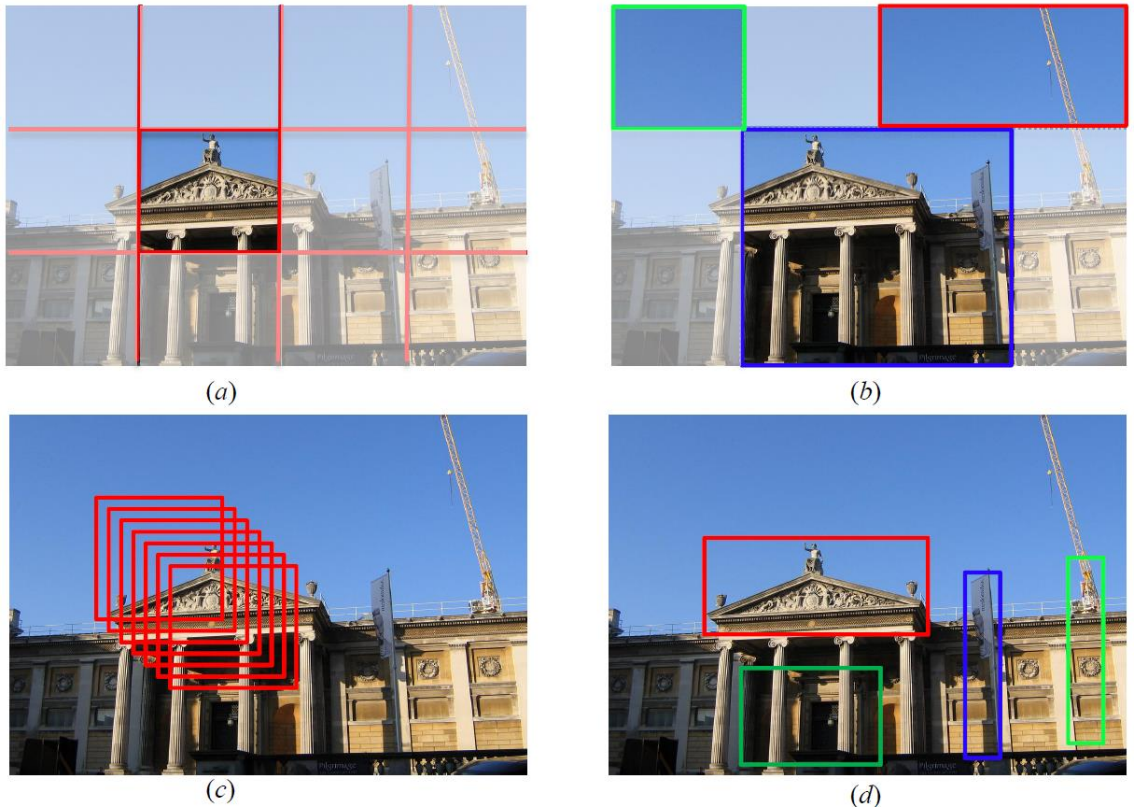


The regions are sampled uniformly with overlaps between consecutive regions



Chen W, Liu Y, Wang W, et al. Deep image retrieval: A survey[J]. arXiv preprint arXiv:2101.11282, 2021.
Tolias G, Sicre R, Jégou H. Particular object retrieval with integral max-pooling of CNN activations[J]. arXiv preprint arXiv:1511.05879, 2015.

Multiple Feed - Forward Pass



Advantage: higher retrieval accuracy

Disadvantage: time-consuming

(a) Rigid grid; (b)Spatial pyramid modeling (SPM); (c) Dense patch sampling;(d) Region proposals (RPs) from region proposal networks.



Feature Embedding

In addition to the direct pooling or regional pooling, we can use embedding to convert the feature maps into compact features. Representative methods include: **BoW, VLAD,** and **FV** .

VLAD generates K visual word centroids, assigns each feature $\vec{x}_t$ to its nearest visual centroid $\vec{c}_k$, and aggregates the difference $(\vec{x}_t, \vec{c}_k)$ as

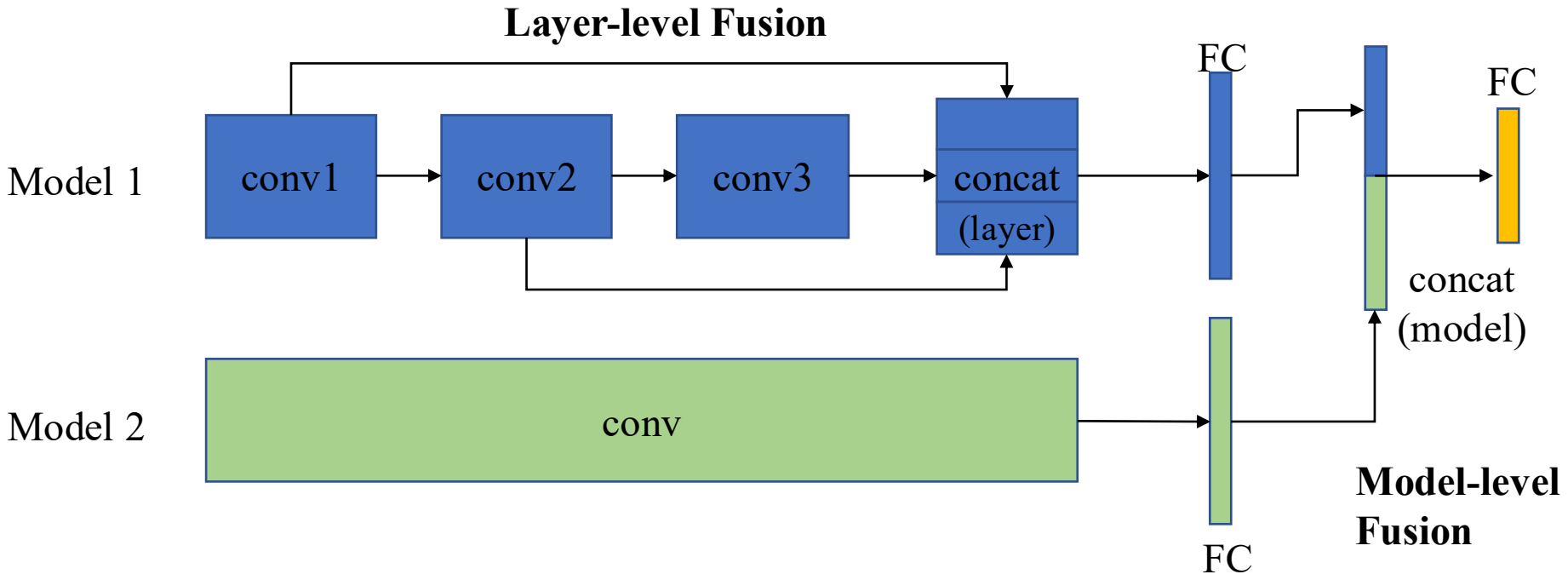
$$g(\vec{c}_k) = \frac{1}{T}\sum_{t=1}^{T} \phi(\vec{x}_t, \vec{c}_k)(\vec{x}_t - \vec{c}_k)$$

$$\phi(\vec{x}_t, \vec{c}_k) = \begin{cases} 1, if\ \vec{c}_k\ is\ the\ nearest\ codeword\ for\ \vec{x}_t \\ 0, therwise \end{cases}$$

The VLAD representation is stacked with the residuals to all centroids, with dimension $(D \times K)$:

$$G_{VLAD}(\vec{x}) = [\ldots, g(\vec{c}_k)^{\intercal}, \ldots]^{\intercal}$$

Feature Fusion



Chen W, Liu Y, Wang W, et al. Deep image retrieval: A survey[J]. arXiv preprint arXiv:2101.11282, 2021.

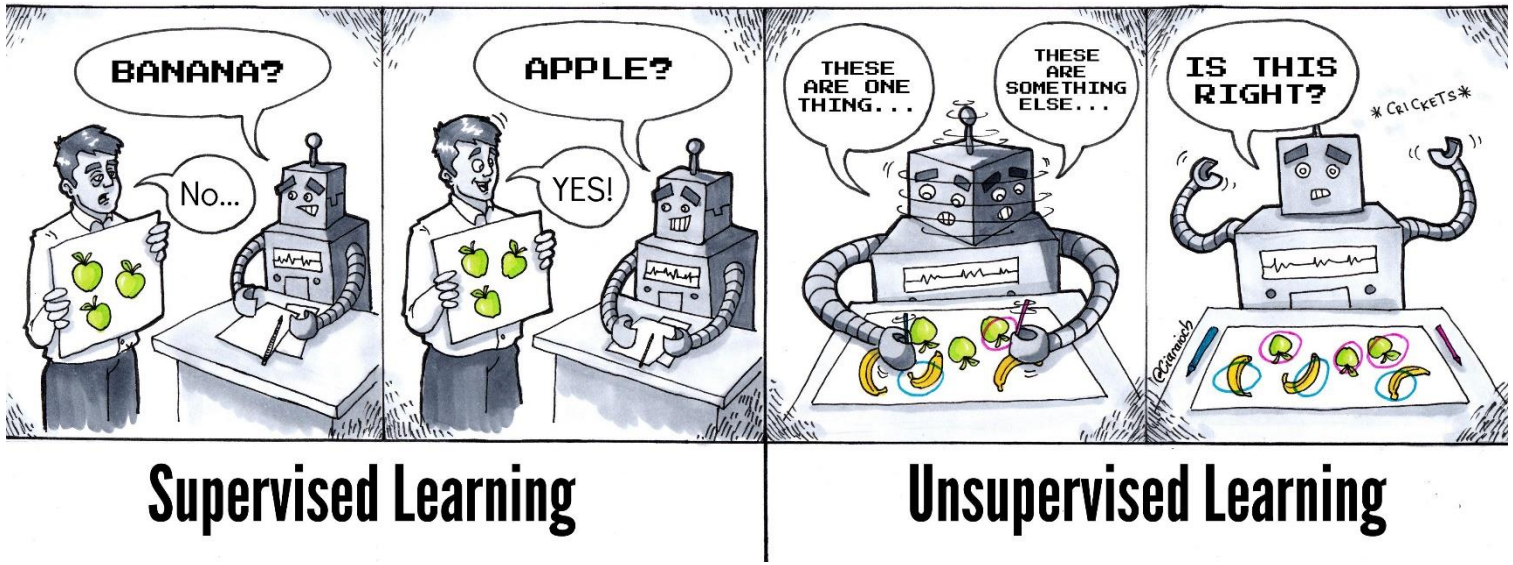


These methods are in fact called **off-the-shell** methods, because they don’t change the parameters (weights) of the original CNNs.



By contrast, there are **fine-tuned methods**, in which we can update the parameters (weights) for better performance (to address the **domain shift**).

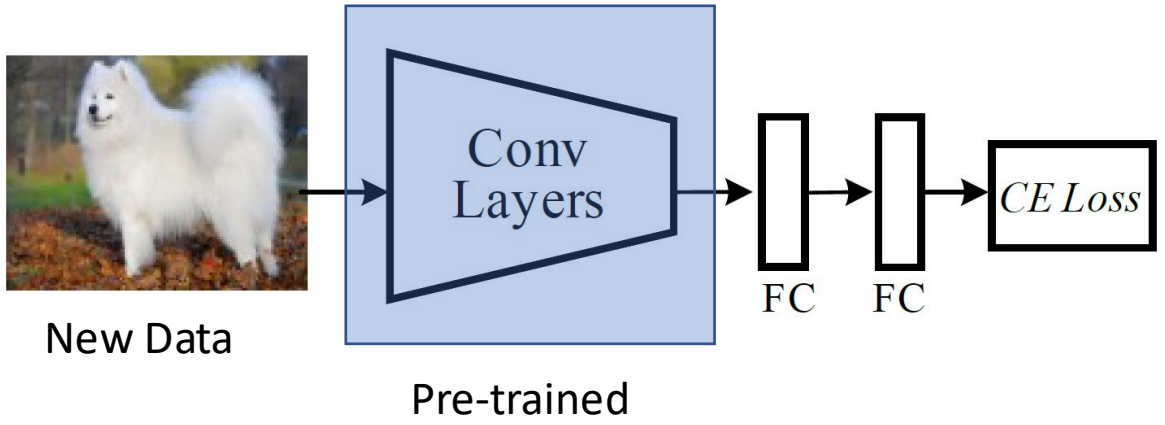
Fine-Tuning



https://twitter.com/athena_schools/status/1063013435779223553

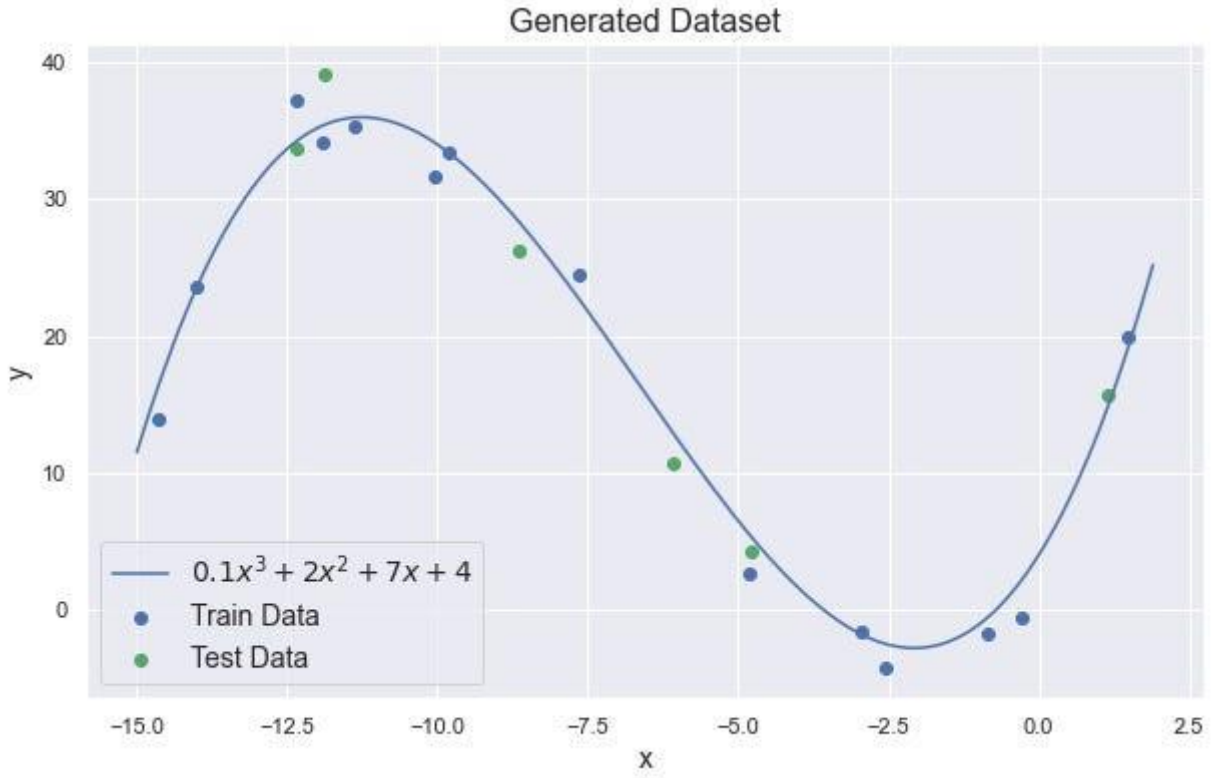
Classification-based Tuning

Retrain the pre-trained DCNN (AlexNet, VGG, GoogLeNet, or ResNet) when labels on the new datasets are available so as to improve the model-level adaptability on the new datasets.



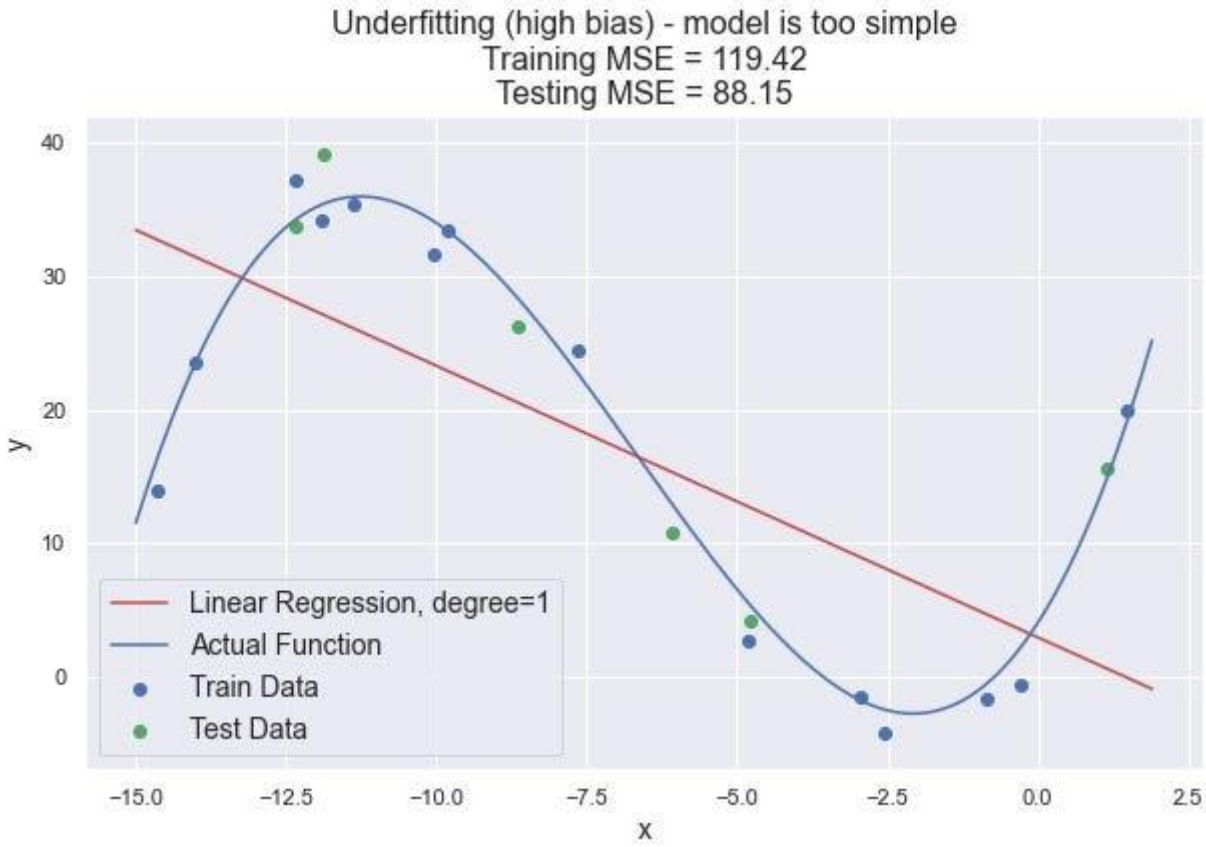
Chen W, Liu Y, Wang W, et al. Deep image retrieval: A survey[J]. arXiv preprint arXiv:2101.11282, 2021.

Overfitting and Underfitting



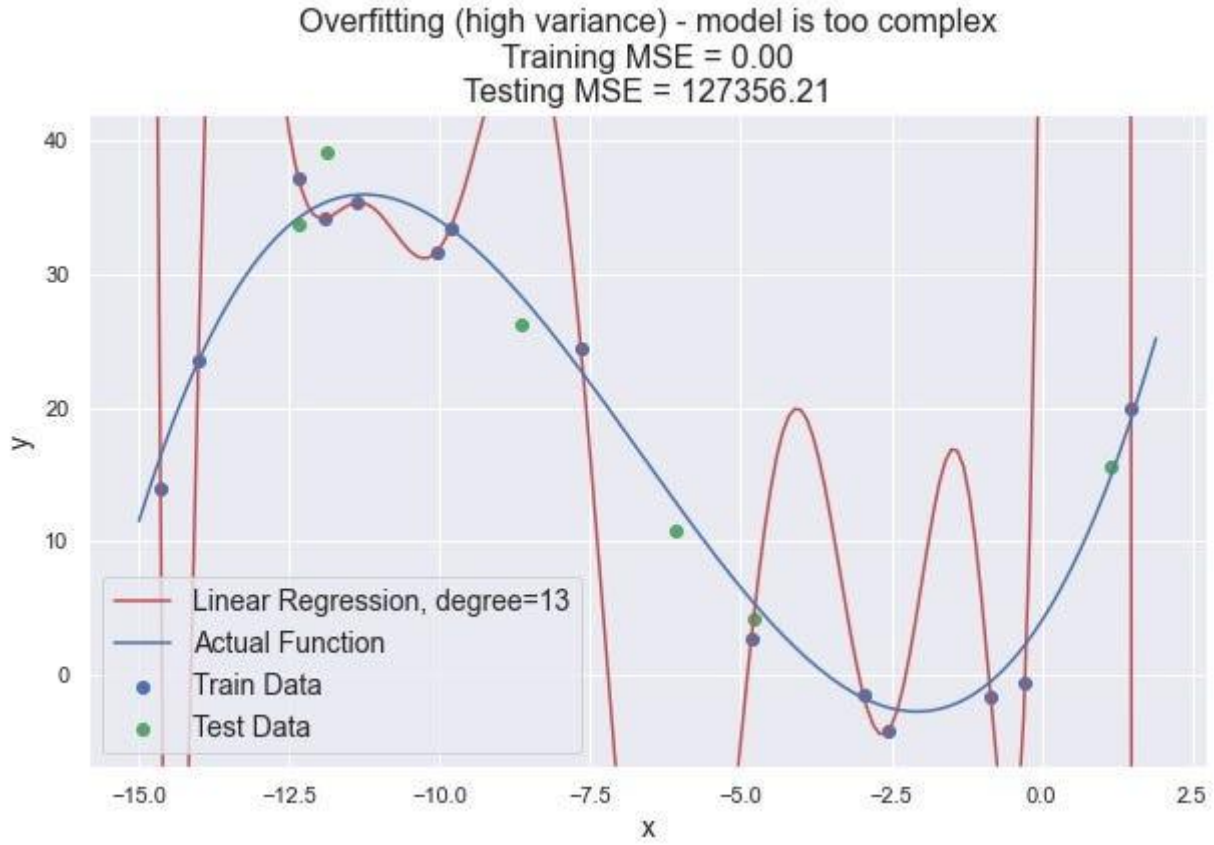
https://towardsdatascience.com/overfitting-and-underfitting-principles-ea8964d9c45c

Overfitting and Underfitting



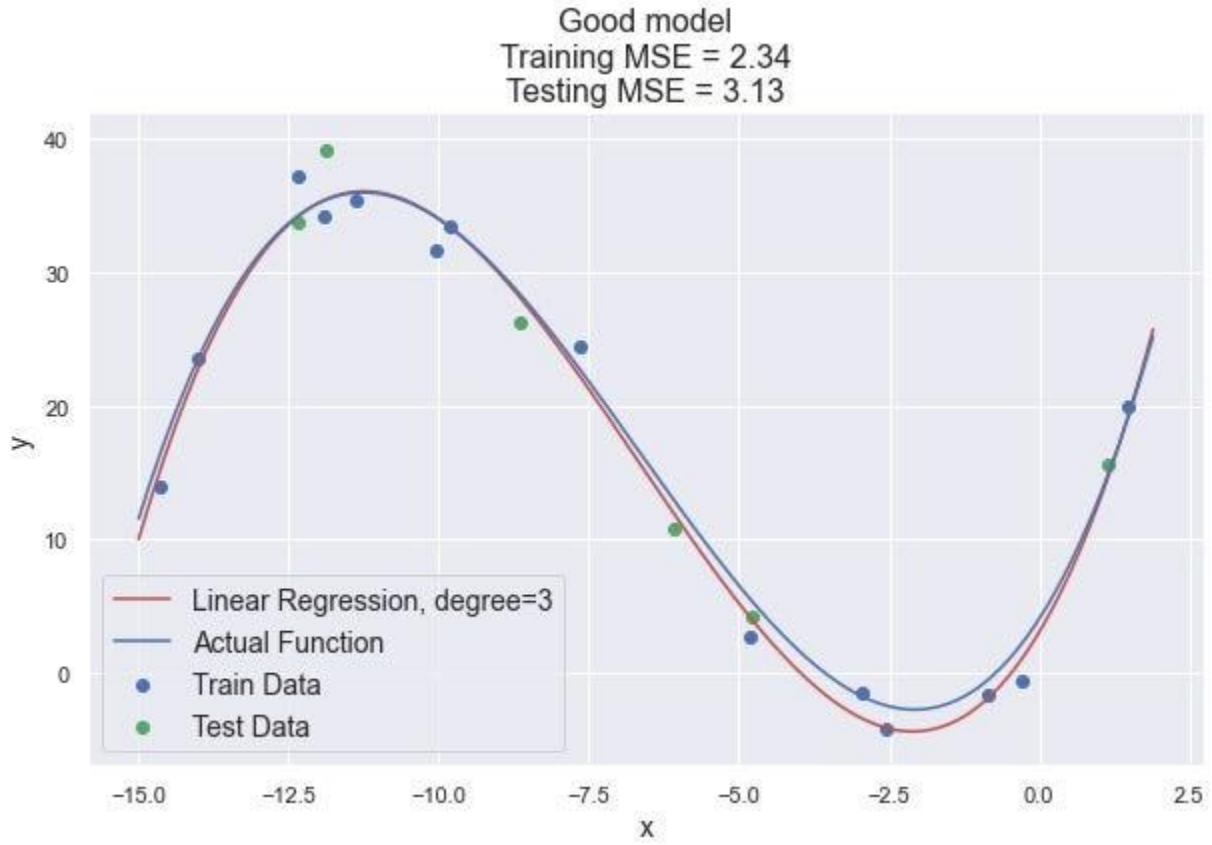
https://towardsdatascience.com/overfitting-and-underfitting-principles-ea8964d9c45c

Overfitting and Underfitting



https://towardsdatascience.com/overfitting-and-underfitting-principles-ea8964d9c45c

Overfitting and Underfitting

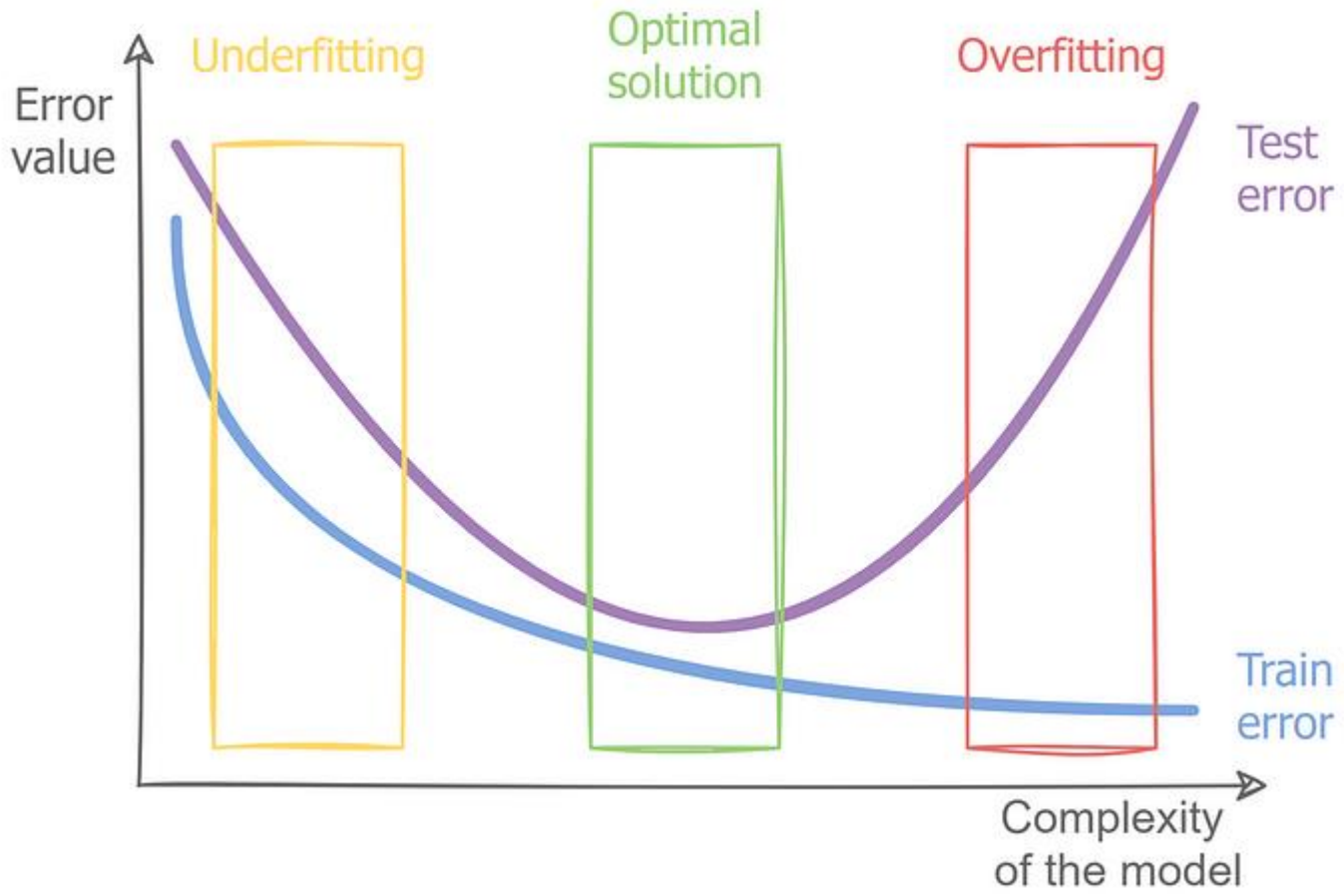


https://towardsdatascience.com/overfitting-and-underfitting-principles-ea8964d9c45c



How do we know?

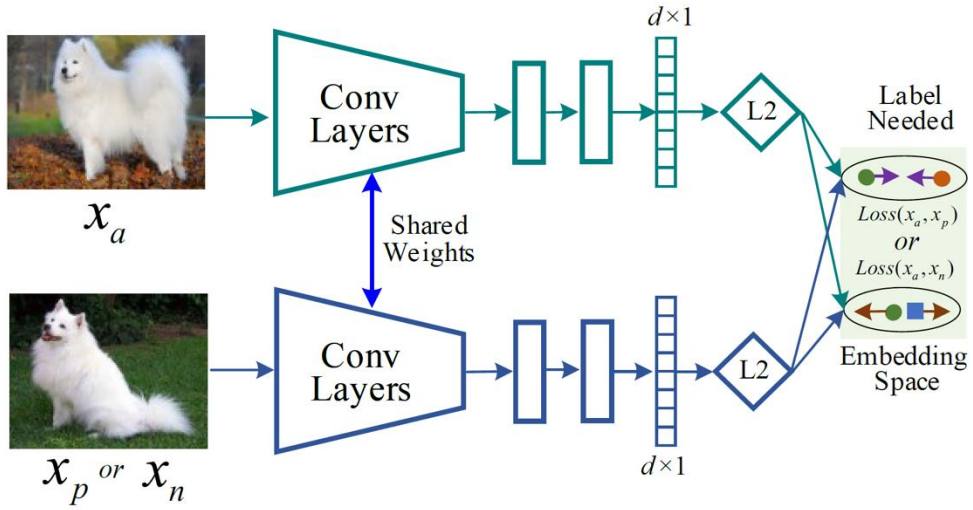
Overfitting and Underfitting



https://towardsdatascience.com/overfitting-and-underfitting-principles-ea8964d9c45c

Verification-based Tuning

1) A pair-wise constraint (e.g., Siamese network)

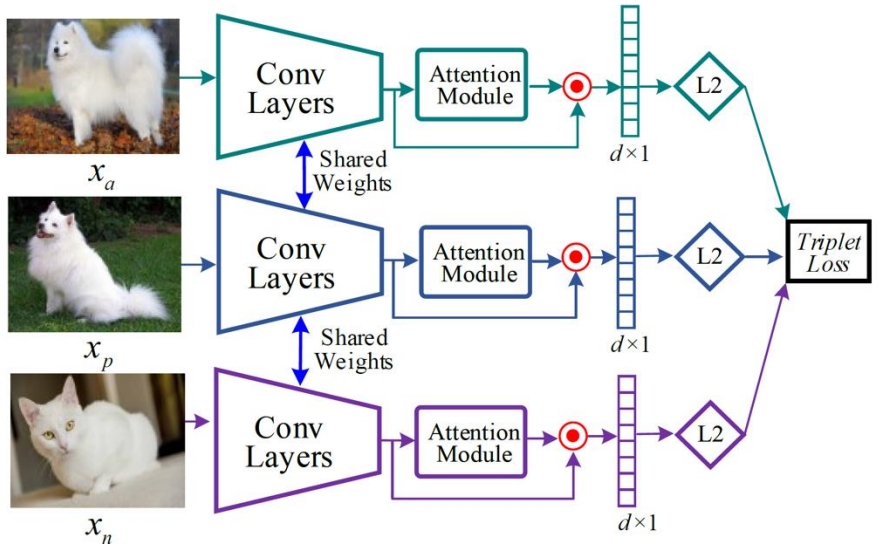


$$L_{Siam}(x_i, x_j) = \frac{1}{2} S(x_i, x_j) D(x_i, x_j) + \frac{1}{2}(1 - S(x_i, x_j)) \max(0,\ m - D(x_i, x_j))$$

$$D(x_i, x_j) = ||f(x_i; \boldsymbol{\theta}) - f(x_j; \boldsymbol{\theta})||_2^2 \qquad S(x_i, x_j) \in \{0, 1\}$$

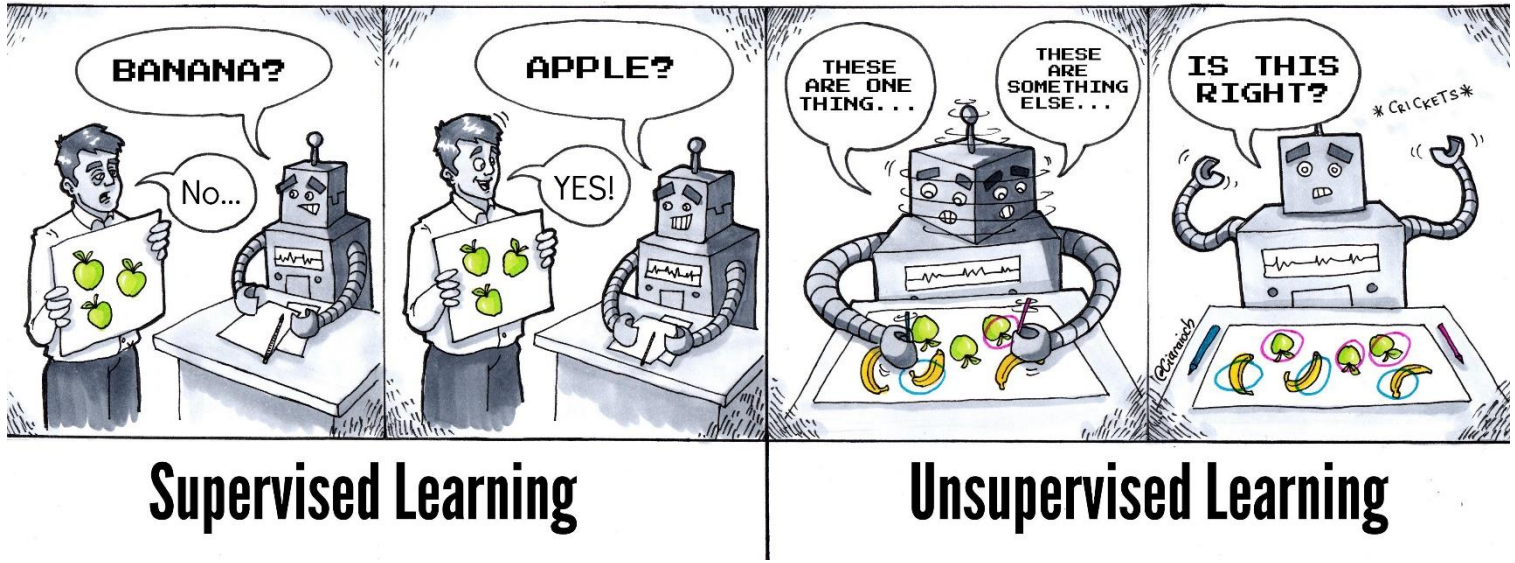
Verification-based Tuning

2) A triplet constraint (e.g., triplet networks)



$$L_{Triplet}(x_a, x_p, x_n) = \max(0, m + D(x_a, x_p) - D(x_a, x_n)))$$

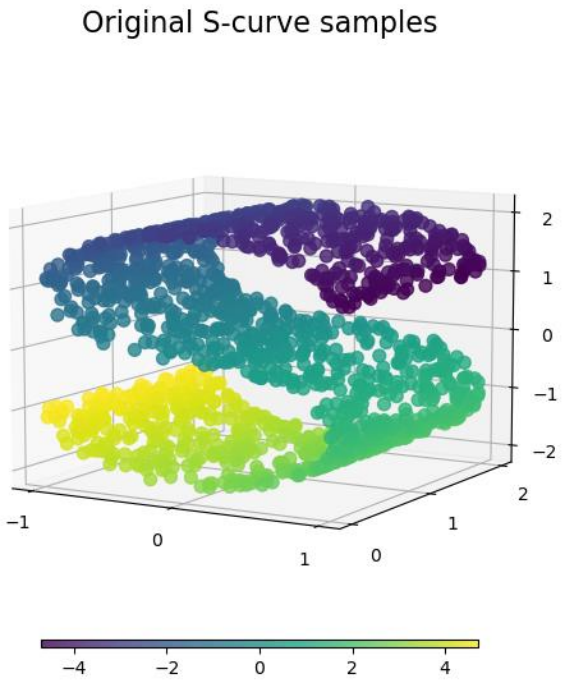
Fine -Tuning

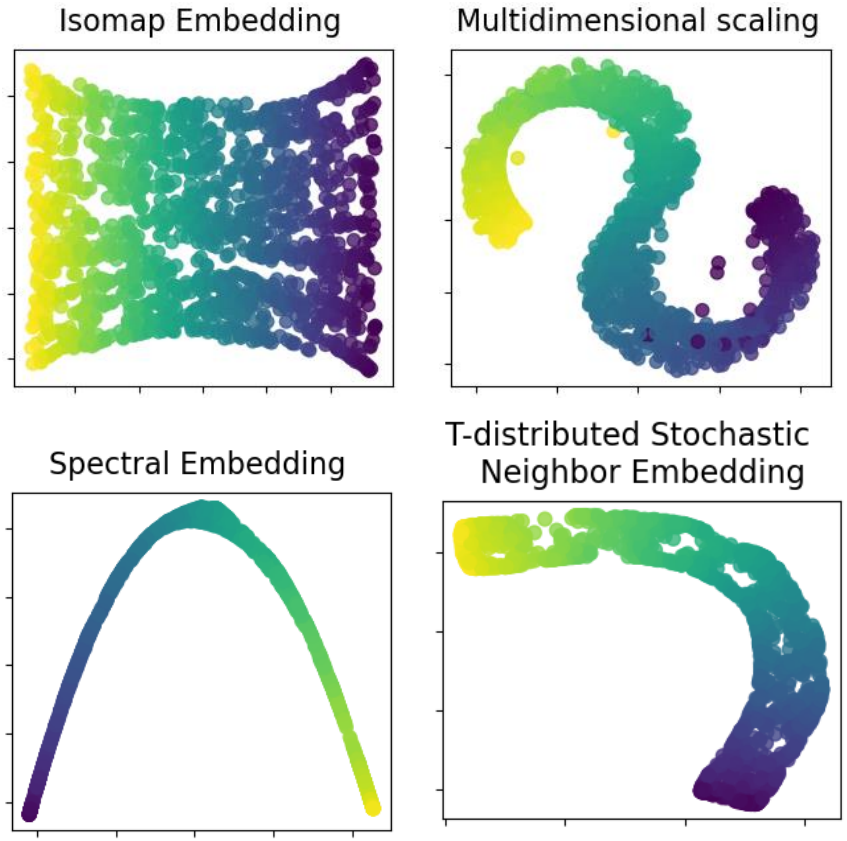


https://twitter.com/athena_schools/status/1063013435779223553

Unsupervised Tuning

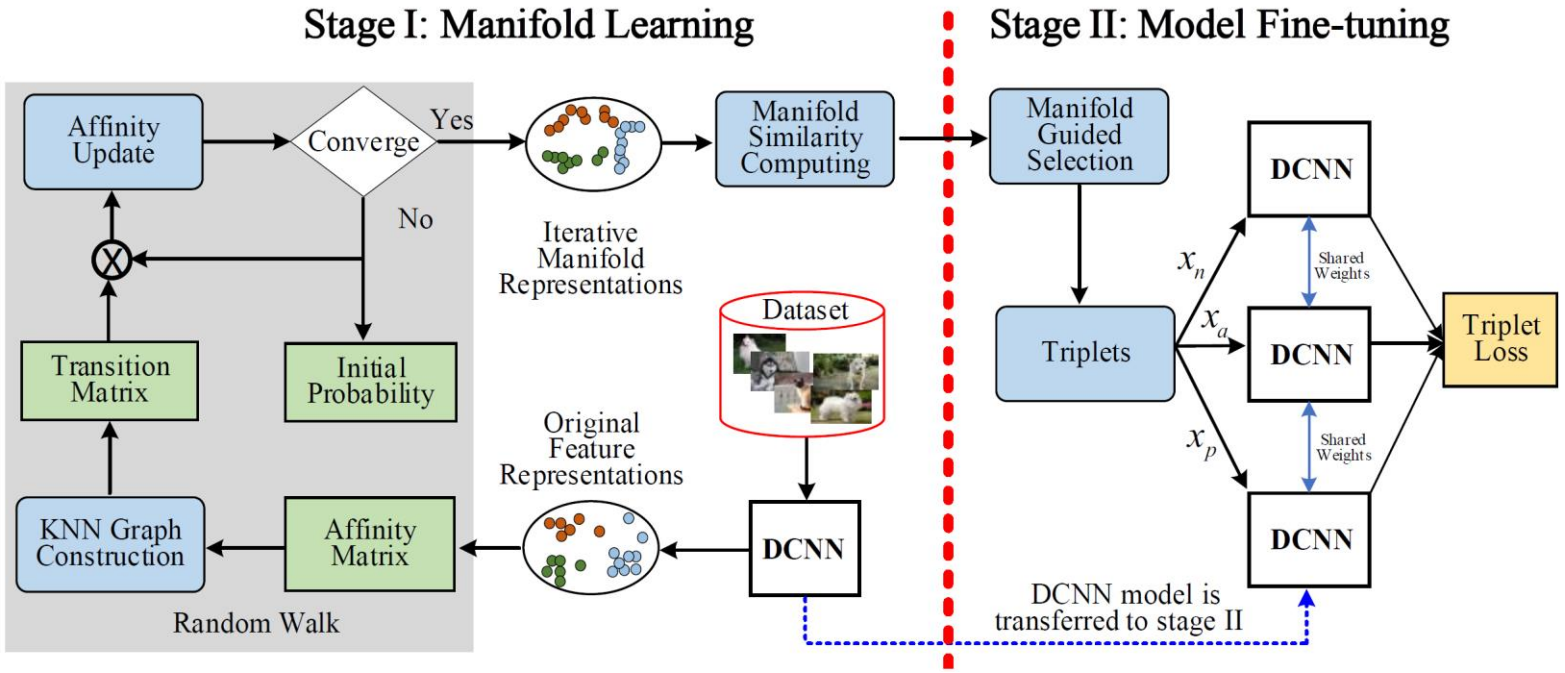
Manifold learning is a method for non-linear dimensionality reduction, which learns the intrinsic correlation of data in a high dimensional space and represent them in a low dimensional space (with the correlation preserved).



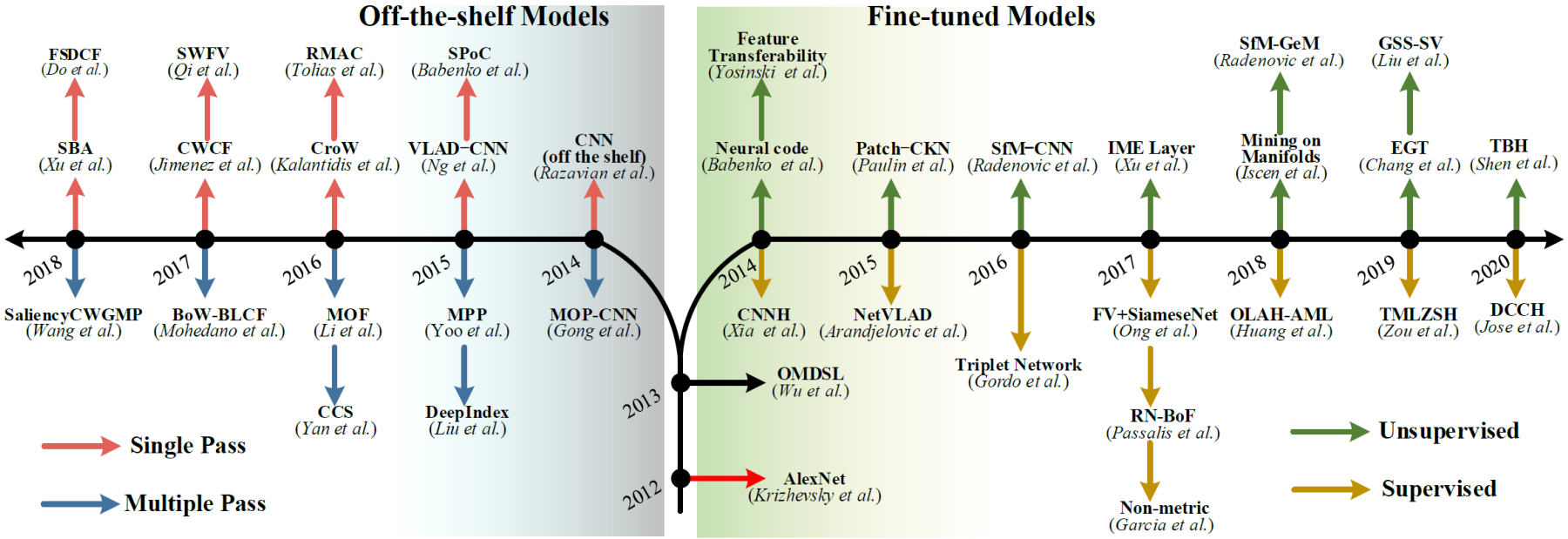


Unsupervised Tuning

Manifold learning is used as a way to guide the sampling of positive and negative pairs.



Deep Image Retrieval



Chen W, Liu Y, Wang W, et al. Deep image retrieval: A survey[J]. arXiv preprint arXiv:2101.11282, 2021.

After-class Reflection and Beyond

1. **How is image retrieval different from image classification?**
   In your answer, compare the input, output, and goal of a CNN-based classification system and a CNN-based retrieval system.

2. **Why can the feature extraction part of a CNN be useful even if we remove the final classification layer?**
   Explain how CNN features can be used to represent images for retrieval.

3. **What are feature maps in a CNN, and why are they useful for image retrieval?**
   Discuss how each feature map can be interpreted as a response to some learned visual pattern.

4. **How do low-level, mid-level, and high-level CNN features differ from each other?**
   Give examples of what early, middle, and deeper layers may capture in an image.

5. **Why might directly flattening raw CNN feature maps not be the best representation for image retrieval?**
   Discuss possible issues such as high dimensionality, redundancy, and lack of compactness.

After-class Reflection and Beyond

6. **What is the main idea of R-MAC?**
   Explain how regional maximum activations of convolutions use multiple image regions to form one compact descriptor.

7. **What are the advantages and disadvantages of multiple feed-forward pass methods?**
   Discuss the trade-off between retrieval accuracy and computational cost.

8. **How is VLAD different from Bag of Visual Words?**
   Explain why aggregating residuals relative to visual word centers may preserve more information than simply counting visual word occurrences.

9. **What is domain shift, and why is it a problem for off-the-shelf retrieval methods?**
   Give an example where a CNN trained on ImageNet may not be ideal for a new retrieval task.

10. **What is fine-tuning, and how can it improve image retrieval performance?**
    Explain how updating the parameters of a pre-trained CNN can adapt the feature space to a new domain.

11. **Why might verification-based tuning be more directly related to image retrieval than classification-based tuning?**
    Compare training a network to classify individual images with training it to judge whether two images are similar.

Thank You!

